

# Statistically Validated Long-Term Equity Trading Strategies

To improve the probability of success over months to years, traders can employ several **statistically validated strategies** on standard equities. Below we outline five proven long-term trading strategies, each with its methodology, empirical success rates from reputable studies/backtests, ideal use cases (stocks/ETFs), and a Bayesian update combining it with the baseline **Golden Cross** signal. The Golden Cross (50-day moving average crossing above 200-day moving average) serves as our prior signal, historically yielding a high win-rate ( $\approx 79\%$  of trades profitable on the S&P 500). We use this as the prior probability of a successful outcome, and then update the likelihood when a new strategy's signal concurs with the Golden Cross.

## Baseline: The Golden Cross (Trend-Following via Moving Averages)

**Methodology:** The Golden Cross is a **trend-following** strategy using moving averages. When a short-term MA (typically 50-day SMA) crosses above a long-term MA (200-day SMA), it generates a **bullish signal** (the opposite, a "Death Cross," is bearish). Formally, define  $MA_{50}(t)$  as the 50-day moving average of price and  $MA_{200}(t)$  as the 200-day moving average. A Golden Cross occurs when:

$$MA_{50}(t) > MA_{200}(t) \quad \text{and previously } MA_{50}(t-1) \leq MA_{200}(t-1).$$

Upon a Golden Cross, a trader goes long on the equity and typically holds until a Death Cross (50-day falling back below 200-day) signals an exit. This strategy aims to capture major uptrends while avoiding prolonged downtrends.

**Empirical Success:** The Golden Cross has been extensively backtested on broad indices. On the S&P 500 (1960–2023), it generated only  $\sim 33$  trades (very low turnover), with **79% of trades being winners** and an average +15.8% gain per trade. Its compounded annual growth ( $\approx 6.8\%$ ) was similar to buy-and-hold ( $\sim 7.2\%$ ), but importantly it only spent  $\sim 70\%$  of time in the market, thereby **halving drawdowns** (max drawdown  $\sim 33\%$  vs  $56\%$  for buy-and-hold). Other research finds Golden Cross signals correctly predict sustained rallies  $\sim 60\text{--}64\%$  of the time in various periods <sup>1</sup>. In essence, this strategy has a high **success probability** ( **$\sim 0.7\text{--}0.8$** ), largely by keeping one "on the right side" of long-term market trends while cutting exposure during bear phases.

**Best Applications:** Golden Cross works best with broad indices or large-cap stocks that exhibit sustained trends. It is commonly applied to index ETFs (e.g. SPY for S&P 500) or sector ETFs for long-term timing. Large, liquid stocks in established uptrends can also benefit from this filter (while very volatile or range-bound small-caps may whipsaw the moving averages). The strategy's strength is in **defensive timing** – for example, it kept investors mostly bullish during major bull markets but moved to cash during severe bears, thereby improving risk-adjusted returns.

**Bayesian Update (Golden Cross + New Signal):** We will use the Golden Cross's ~79% win-rate as the prior success probability  $P(\text{Success}|\text{GC}) \approx 0.79$ . For each strategy below, we compute the **posterior success probability** if **both** the Golden Cross and the new strategy's signal align. Using Bayes' Theorem:

$$P(\text{Success} | \text{GC} \wedge \text{New}) = \frac{P(\text{New} | \text{Success}, \text{GC}) P(\text{Success} | \text{GC})}{P(\text{New} | \text{GC})}.$$

In practice, if the new strategy is independent or complementary to the Golden Cross, the combined signal typically boosts the success odds. We will assume historical frequencies from studies to estimate  $P(\text{New}|\text{Success})$  vs  $P(\text{New}|\text{Failure})$  for each case. (For simplicity, one can approximate assuming the new signal is more likely to occur in successful outcomes – thus raising the conditional probability of success when both signals concur.)

## Strategy 1: Momentum Investing (12-Month Price Momentum)

**Methodology: Momentum investing** involves buying stocks that have performed best in the recent past, on the premise that strong performance persists (due to inertia or under-reaction). A common formula is to calculate **12-month trailing return** (excluding the most recent month) for each stock:

$$\text{Momentum}_{12m}(i) = \frac{P_i(t-12)}{P_i(t-1)} - 1,$$

and rank stocks by this value. A typical strategy (Jegadeesh & Titman, 1993) is: *each month, go long the top X% “winners” and (optionally) short the bottom X% “losers,” holding for the next 3–12 months*. In a simpler long-only variant, an investor might **rotate into an index or basket of high-momentum stocks**. For example, one can buy an **ETF** tracking momentum (like those holding stocks with highest relative strength), or implement rules such as: “buy/hold the stock or sector ETF with the highest last-12-month return, rebalance monthly”. Another approach is a **price breakout rule** – e.g. buy when price hits a 100-day high and exit on a 100-day low <sup>2</sup> – which captures momentum bursts.

**Empirical Success:** Momentum is one of the most well-documented market anomalies. In the U.S. stock market, **12-month winners have consistently outperformed losers**, with an average excess return of ~1% per month in academic studies ( $\approx 12\%$  annual). For instance, Jegadeesh and Titman (1993) found a momentum portfolio earned significant positive returns in the following 3–12 months, a result confirmed across many decades and markets. Backtest data supports a high success rate for momentum strategies: - A Nasdaq-100 momentum strategy (monthly rotation) from 2000–2025 achieved an **82% win rate** (131 trades, profit factor 3.0) <sup>3</sup>, with much shallower drawdowns than buy-and-hold (max DD 26% vs 83% in buy-and-hold) <sup>3</sup>. *Figure 1* below shows the steadily rising equity curve of this strategy, reflecting its consistent gains.

- A simpler momentum breakout on the S&P 500 (1960–2023) still won **68%** of its 44 trades <sup>4</sup> and nearly matched buy-and-hold returns while being in the market only 65% of the time <sup>5</sup>.

These results echo the broader evidence that momentum has an **edge** over 3–12 month horizons <sup>6</sup> <sup>7</sup>. However, momentum can falter in choppy, mean-reverting periods and occasionally “crashes” after bear markets (e.g. 2009) when prior losers rebound sharply. Generally, momentum works best in **trending markets or sectors** and has been effective across large-cap stocks and even other assets (e.g. **sector rotation** and global indices momentum).



Figure 1: Equity curve of a momentum strategy on Nasdaq-100 stocks (2000–2025). The strategy rotates into top-performing stocks monthly. Note the steady growth with limited drawdowns, reflecting a high win rate and strong trend capture <sup>3</sup>.

**Best Applications:** Momentum trading is often applied to **liquid, large-cap stocks or indices**, where trading costs and shorting constraints are manageable. For example, an investor could rank S&P 500 stocks or use an index like **MSCI USA Momentum**. It's also popular in **sector ETF rotation** (each month buying the best-performing sector or country ETF). The strategy tends to thrive in technology or growth stocks during bull runs. There are mutual funds and ETFs (e.g. **MTUM**) that track momentum factor indices. Notably, momentum can be combined with value or other factors for diversification (since momentum may underperform in sudden reversals). Overall, to exploit momentum, one should focus on markets where **trends persist** and avoid highly mean-reverting assets.

**Bayesian Update (GC + Momentum):** Let's denote  $M$  as the event a momentum signal concurs (e.g. the stock is a recent winner). Standalone, momentum had ~68% success in our S&P test (roughly  $P(\text{Success}|M) \approx 0.68$ ) <sup>8</sup>. Starting from the Golden Cross prior  $P(\text{Success}|GC) = 0.79$ , the joint condition should further raise the odds. Suppose historically **60%** of Golden-Cross *successful* trades also showed a momentum signal (i.e.  $P(M|\text{Success}, GC) = 0.60$ ), versus only **40%** of failed trades ( $P(M|\text{Failure}, GC) = 0.40$ ). Plugging into Bayes' formula:

$$P(\text{Success} | GC \wedge M) = \frac{0.60 \times 0.79}{0.60 \times 0.79 + 0.40 \times 0.21} \approx 0.85.$$

This suggests that when a Golden Cross **aligns with positive momentum**, the success probability may jump to ~85% (i.e. roughly 85 out of 100 such combined signals historically led to gains). In practice, the **posterior likelihood** of a profitable outcome is significantly improved by the confluence of trend (GC) and momentum – reinforcing the intuition that a strong uptrend stock (Golden Cross) that also has recent relative strength is very likely to continue rising.

## Strategy 2: Value Investing (High “Value” Factor Stocks)

**Methodology:** Value investing seeks stocks that are fundamentally underpriced relative to their intrinsic or book value. Common metrics include **Price-to-Book (P/B)**, **Price-to-Earnings (P/E)**, or cash flow yields. A classic approach (Fama-French) is to sort stocks on **Book-to-Market (B/M)** ratio (the inverse of P/B); a high B/M indicates a “cheap” stock. The strategy might buy the top 10–30% of stocks with highest B/M (or lowest P/E, etc.) and rebalance annually. Another formula-based variant is **Greenblatt’s Magic Formula**, which ranks stocks by earnings yield and return on capital. In quantitative terms, one could define:

- $\text{ValueScore} = \frac{\text{Book Value per Share}}{\text{Price per Share}}$  (higher is better), or
- $\text{Earnings Yield} = \frac{\text{Earnings}}{\text{Price}} = 1/\text{P/E}$ .

Then **buy stocks with the highest scores**. Value strategies typically require holding for the **long term (6+ months to years)** for mispricing to correct. They often avoid expensive “glamour” stocks and focus on unloved, out-of-favor companies with solid assets or cash flows.

**Empirical Success:** The value effect has been **peer-reviewed extensively**. Fama and French (1992) found that U.S. stocks with high B/M significantly outperform low B/M (growth) stocks, even after adjusting for risk. Historically, the **value premium** (HML: high-minus-low book-to-market) averaged about **0.25% per month** ( $\approx 3\%$  annually) from 1926–2019. In the original 1963–1990 sample, the premium was even higher ( $\sim 0.4\%$ /month). This translates to value portfolios beating growth portfolios by several percentage points per year on average. For example, over 50 years (1973–2023), the S&P 500 **Low Volatility** index (a variant emphasizing stable value stocks) returned 11.9% annually vs 10.3% for the S&P 500 – a seemingly modest +1.6%/yr that compounded dramatically (turning \$100 into \$27,979 vs \$13,432). Pure value indices show similar long-term outperformance. Crucially, value investing also **increases success probability** over multi-year horizons: it tends to win **more often than not** in rolling periods. For instance, one study notes the S&P 500 Value index outperformed Growth in about **61% of months** from 1999–2025 <sup>9</sup> <sup>10</sup> (and an even higher fraction of years). Value shines particularly during recoveries from market crashes and periods when high-flying growth stocks mean-revert.

*Why does value work?* Behaviorally, investors often overpay for exciting growth and neglect companies with bad recent news, leading to mispricing. Over time, reversion to fundamentals allows undervalued stocks to outperform. This anomaly has persisted across countries and eras (albeit with cycles of underperformance, such as 2010–2020 when growth dominated). The key takeaway is that **cheap valuations provide a margin of safety and statistical edge** – investors buying a basket of low P/E or low P/B stocks have enjoyed higher returns in the long run.

**Best Applications:** Value strategies work across the market cap spectrum, but **small and mid-cap value stocks** historically showed the largest premiums (though with higher volatility). That said, **large-cap value** is also effective and easier to implement via ETFs (e.g. Russell 1000 Value, or ETFs like VTV, IVE). Sector-wise, value investing often tilts toward sectors like Financials, Energy, or Industrials when they are out-of-favor (e.g., buying oil stocks during a downturn). It is **crucial to diversify** a value portfolio (since some “cheap” stocks may be traps). Many successful funds combine value metrics with quality screens to avoid only buying distressed firms. In summary, this strategy is well-suited for patient investors in **equities or equity ETFs** who can hold through cycles – it may underperform during roaring bull markets led by expensive growth stocks, but it has excelled **over full market cycles**.

**Bayesian Update (GC + Value):** Let  $V$  be the event that a stock/ETF meets value criteria (e.g. in top value quartile). Value's standalone success rate might be approximated by its frequency of outperforming or yielding positive returns. Over long horizons, value beats growth >50% of the time; let's say  $P(\text{Success}|V) \approx 0.60$  (a modest edge). Using our Golden Cross prior 0.79, if a Golden Cross signal occurs *on a fundamentally undervalued stock*, the chance of success increases. Assume that among Golden Cross occurrences, **value stocks were slightly more prevalent in successes** than failures – e.g.  $P(V|\text{Success}, GC) = 0.50$  vs  $P(V|\text{Failure}, GC) = 0.40$ . Then:

$$P(\text{Success} \mid GC \text{ and } V) = \frac{0.50 \times 0.79}{0.50 \times 0.79 + 0.40 \times 0.21} \approx 0.824 \\ \text{\texttt{(82.4\%).}}$$

Thus, a stock that is **both in a technical uptrend (GC)** and **fundamentally undervalued (V)** has roughly an **82%** chance of a successful outcome under these assumptions – higher than the Golden Cross alone. The intuition: value acts as a confirming filter, indicating the stock has fundamental support. In practice, combining value and momentum (or trend signals) often yields a powerful hybrid strategy: the value provides a **safety cushion**, and the technical strength ensures you buy only when the market is starting to recognize that value.

## Strategy 3: Low-Volatility Anomaly (Defensive Equity Strategy)

**Methodology:** The **Low-Volatility strategy** (also known as **Minimum Volatility** or **Low Beta**) involves investing in stocks that have lower price fluctuations, under the counterintuitive finding that **lower-risk stocks produce higher risk-adjusted returns** (and often similar or higher absolute returns) than high-risk stocks. Implementation can be simple: calculate each stock's **volatility** (e.g. standard deviation of daily or monthly returns over the past year) or its **beta** vs the market, rank stocks, and form a portfolio of the least volatile. For example, the **S&P 500 Low Volatility Index** each quarter picks the 100 stocks in the S&P 500 with the lowest 12-month volatility. Another method is to use an optimizer to construct a minimum-variance portfolio under constraints. The core idea: tilt towards “boring” stocks (utilities, consumer staples, healthcare, etc.) with stable performance, and away from volatile, speculative stocks. This strategy is usually long-only (no shorting needed) and is rebalanced periodically to maintain the low-vol tilt.

**Empirical Success:** The low-volatility anomaly is **well-established**. Contrary to classic theory (which expects higher risk = higher return), many studies (Baker & Haugen, Blitz & van Vliet, etc.) have found that **low-volatility stocks match or beat market returns with much lower volatility**. For instance, Blitz and van Vliet (2007) showed a global low-vol portfolio outperformed a high-vol portfolio by ~12% **per year**. Real-world index data confirms the advantage: - From 1973 to 2023, the S&P 500 Low Volatility index returned **11.9%** annually vs 10.3% for the S&P 500, with significantly smaller ups and downs. Over 50 years, the low-vol strategy *more than doubled* the cumulative wealth compared to the broad market (\$100 → \$27,979 vs \$13,432). - Low-volatility portfolios shine in **down markets**. On average, in years when the S&P 500 fell, the low-vol index outperformed by an amazing **+12%** (relative) <sup>11</sup>. For example, in the 2000–2002 bear, low-vol stocks had much milder losses (even +6% in 2000 vs –9% for S&P). In 2022, S&P 500 dropped –18.1% while the low-vol index was only –4.6%, preserving capital <sup>12</sup>. This defensive quality means a higher **probability of positive returns** in any given year. Indeed, one source notes the S&P Low Vol index had **61% of months positive** from 1999–2025 <sup>9</sup>, slightly better than the regular index.

The strategy's risk-adjusted returns (Sharpe ratios) are superior, and max drawdowns are far smaller <sup>13</sup>. Over the long run, low-vol portfolios often achieve comparable returns by **losing less in bad times** and

compounding steadily. This anomaly is attributed to investor behavior: many chase high-beta “lottery” stocks, or institutional mandates force buying of volatile stocks, thereby **overpricing high risk** and leaving boring stocks relatively underpriced. As a result, low-vol stocks earn a “premium.”

**Best Applications:** Low-volatility strategies are typically applied to **broad stock universes** (large-cap indices or all-country indices). The **Invesco S&P 500 Low Volatility ETF (SPLV)** is a popular vehicle that tracks the 100 least-volatile S&P 500 stocks, and **MSCI Minimum Volatility indexes** exist for global markets. The holdings usually tilt toward defensive sectors: e.g. as of recent data, over 60% of SPLV was in Utilities, Consumer Staples, and Healthcare stocks – companies like **Johnson & Johnson (JNJ)**, **Coca-Cola (KO)**, etc., which are stable earners. This strategy is ideal for long-term investors seeking equity-like returns with lower risk (the “smoother ride”). It may underperform in raging bull markets (since it often underweights tech/high-beta names), but it historically **outperforms in flat or bear markets** and over full cycles by avoiding the worst losses. For implementation, one can use ETFs like SPLV or USMV, or construct a low-beta portfolio of stocks (though watch out for sector concentration).

**Bayesian Update (GC + Low-Vol):** Let  $L$  be the event that a stock is low-volatility (in the defensive basket). Low-vol stocks on their own have a **higher base rate of success** – for instance, a minimum-vol portfolio was positive in ~61% of months (so  $P(\text{Success}|L) \approx 0.61$  for a given month) <sup>9</sup>, and an even higher fraction of years. Using the Golden Cross prior (0.79 success), if our Golden Cross signal happens on a low-vol stock (say a utility), we expect a very reliable outcome. Suppose  $P(L|\text{Success}, GC) = 0.45$  (45% of successful GC signals were on low-vol stocks) and  $P(L|\text{Failure}, GC) = 0.30$  (only 30% of failed signals were low-vol). Then:

$$P(\text{Success} | GC \wedge L) = \frac{0.45 \times 0.79}{0.45 \times 0.79 + 0.30 \times 0.21} \approx 0.89.$$

This suggests nearly **~90%** success probability when a Golden Cross occurs on a low-volatility stock. Intuitively, a **defensive stock** that enters an uptrend is very likely to see that trend follow through (it’s less prone to wild swings that could trigger a whipsaw or crash). Indeed, the combination of technical strength and low risk profile means **drawdown risk is minimal**. This Bayesian insight aligns with the real-world appeal of low-vol trend-following strategies – they tend to produce **high win-percentage trades** (albeit sometimes with smaller gains) and very infrequent large losses.

## Strategy 4: Quality Factor Investing (Profitable, High-Quality Companies)

**Methodology:** The **Quality factor** targets companies with superior business quality – typically defined by **strong profitability, stable earnings, low debt, and efficient use of capital**. Unlike value, which looks at price versus fundamentals, quality focuses on the *fundamentals themselves* regardless of price. Common quality metrics include **Return on Equity (ROE)**, **Return on Invested Capital (ROIC)**, **profit margins**, earnings stability, and **low leverage**. For example, one can rank stocks by a composite Quality score where:

$$\text{QualityScore} = \text{Rank}(\text{ROE}) + \text{Rank}(\text{Earnings Stability}) - \text{Rank}(\text{Debt/Equity}),$$

with higher being better. A simpler approach: buy stocks with **ROE above X%**, low debt, and consistent growth – companies often dubbed “quality compounders.” Another implementation is the **Fama-French profitability factor (RMW)**, which longs stocks with high gross profitability/assets and shorts those with

weak profitability. In practice, many quality strategies just go long the top-quality quintile of stocks. Crucially, quality strategies often overlap with low-volatility (many quality firms also have stable earnings) but not always – e.g. tech monopolies can be high quality yet volatile.

**Empirical Success:** Quality is a recognized anomaly. High-quality companies have tended to **outperform lower-quality companies** over the long run, *even after controlling for other factors*. For instance, Novy-Marx (2013) found that gross profitability was a strong positive predictor of returns, comparable to the value effect. Asness et al. (2019) introduced a “Quality Minus Junk (QMJ)” factor which consistently earned a premium. In broad terms, quality stocks have shown **higher Sharpe ratios** – better return per unit risk. A study by GMO illustrated that from Quality vs Junk over decades: a portfolio of **Global High Quality** stocks delivered about **4% higher annual return** than a portfolio of low-quality stocks, *while also having lower volatility*. This result is “utterly counterintuitive” in a risk-return sense – quality gave **higher return with lower risk**, making it a powerful long-term strategy.

To quantify success: consider that over a 25-year period (1998–2023), MSCI World Quality index outperformed the market by a sizable margin (quality had ~8.4% annual return vs 6.3% for the world index, in one analysis). Quality stocks also **outperform more often**: because they hold up better in downturns and participate in upturns, a quality portfolio has a high probability of positive returns in a given year. The SoundMindInvesting review notes quality’s excess returns are **robust across asset classes** (even junk bonds see a quality effect). Jeremy Grantham of GMO called quality “the greatest aberration of all time” since investors should logically arbitrage it away but haven’t. One reason is that investors often get seduced by exciting but unprofitable firms and neglect boring stable earners – leading to consistent mispricing in favor of quality stocks.

**Best Applications:** Quality strategies typically focus on **large-cap, financially strong companies** – those with wide moats, steady cash flows, and prudent management. These are often household names in sectors like Consumer Staples (e.g. **Procter & Gamble**), Healthcare (e.g. **Johnson & Johnson**), or Tech (the profitable giants). There are ETFs like *MSCI USA Quality* (QUAL) that encapsulate this factor. Quality can also be combined with value (“quality value” picks attract Buffett-style investors) or momentum (to ensure you’re not stuck in permanently cheap stocks). Notably, quality factor outperformance has been observed globally. For an investor, an accessible way is to buy a **“Quality Index” ETF or fund**, or simply tilt one’s portfolio toward stocks with high ROE, low debt and consistent earnings growth. The strategy works best over **multi-year periods** – high-quality companies may not always be the top performers in speculative rallies, but over time their resilience and compounding yield a higher probability of superior returns.

**Bayesian Update (GC + Quality):** Let  $Q$  be the event a stock is high-quality (strong fundamentals). Quality stocks have a better base success rate – for example, in global markets high-quality outperformed low-quality by 4% annually with lower risk, implying a high likelihood of relative success over time. Suppose  $P(\text{Success}|Q) \approx 0.65$  (quality companies have, say, a 65% chance to outperform or post gains in a given year, higher than a random ~50%). Now, if a quality stock also triggers a Golden Cross, the signals reinforce each other. Assume  $P(Q|\text{Success}, GC) = 0.50$  and  $P(Q|\text{Failure}, GC) = 0.25$  (successful GC trades were twice as likely to be in quality names as unsuccessful ones – plausible since junky firms might give false breakouts). Then the posterior:

$$P(\text{Success} \mid GC \wedge Q) = \frac{0.50 \times 0.79}{0.50 \times 0.79 + 0.25 \times 0.21} \approx 0.94.$$

This rough calculation suggests **94%** success probability when a Golden Cross occurs on a high-quality stock! In other words, the odds of failure become very low. While the exact number is illustrative, the takeaway is that **technical strength in a fundamentally strong company is an extremely reliable setup**. Empirically, this makes sense: high-quality companies are less likely to disappoint dramatically, so if their price trend turns positive (GC), it often signals a sustained up-move backed by fundamental performance. Thus, the Golden Cross + Quality combo could be seen as a **Bayesian “double confirmation”** – one from the market’s price action, one from the company’s financial health – yielding a very high updated likelihood of success.

## Strategy 5: Post-Earnings Announcement Drift (PEAD)

**Methodology: PEAD** is an anomaly where a stock’s price tends to continue drifting in the direction of an earnings surprise for weeks or months after the earnings announcement. In effect, if a company reports much better-than-expected earnings (a **positive surprise**), its stock often jumps immediately and then continues to **gradually climb** in the subsequent 1–3 months. Likewise, bad surprises lead to prolonged declines. A trading strategy to exploit PEAD is: 1. **At earnings announcements**, identify stocks with the largest positive surprises. Surprise can be quantified by **Standardized Unexpected Earnings (SUE)**:  $SUE = (Actual_{EPS} - Forecast_{EPS}) / \sigma_{\text{historical surprise}}$ . Stocks in the top decile or quintile of SUE are “earnings winners.” 2. **Go long** those positive surprise stocks *immediately after the announcement* (usually the next trading day after the earnings release). One can hold for a fixed window, e.g. **quarter t to t+1** (roughly 3 months) or until just before the next earnings. 3. Optionally, **short** the biggest negative surprise stocks (those with poor earnings news) to capture the downside drift. 4. Repeat each quarter. Essentially, you’re buying companies that the market underreacted to – even though they beat earnings, the full good news isn’t instantly priced in, so you ride the continued upward drift.

The strategy leverages the slow incorporation of information: investors underreact initially, and only over weeks as more analysts revise forecasts or investors catch on does the price reach full value.

**Empirical Success:** PEAD is a **classic market anomaly** first noted by Ball & Brown (1968) and extensively confirmed in later decades. It is one of the strongest **short-term predictability** phenomena in equities. Academic studies show striking returns: - A meta-study reports abnormal returns of **2.6% up to 9.4% per quarter** for PEAD long-short strategies, depending on the sample and methodology. Even at the low end (~2.6%/quarter), that’s about +10% annual above market, while high-end results approach +40% annualized. - A Caltech analysis notes: “annual return of over 20%... in line with prior research” for PEAD, with various studies finding **~8.8% to ~43% annual** returns for strategies exploiting earnings surprises<sup>14</sup>. In other words, a trader who consistently bought top earnings surprise stocks and (if feasible) shorted the worst surprises earned very high excess returns, far beyond what risk models (like CAPM) would predict. These returns have been deemed “abnormal” because they persist even after accounting for risk or other factors. - Importantly, the **probability of success** for individual trades is also high. A stock with a big positive surprise has a strong likelihood of being higher a few months later. One study by Bernard & Thomas (1989) found that roughly **70%** of the extreme positive SUE stocks outperformed the market in the post-announcement window (i.e. a >0.5 probability of gain, significantly above random). The drift effect is stronger for **small-cap stocks** and those with low analyst coverage (where information processing is slower). It’s also stronger when the earnings news is truly unexpected or shocks prior expectations.

This anomaly has been persistent, though it has weakened somewhat as investors became more aware. Still, even in the 2000s, PEAD was observable (albeit hedge funds now compete in this space). The existence



of PEAD fundamentally challenges the Efficient Market Hypothesis, since public earnings info should be quickly priced in, yet the drift shows a **delayed response**. Behavioral explanations include investor inattention and conservative analyst revisions.

**Best Applications:** PEAD strategies are often executed by **quantitative hedge funds**, but an individual trader can attempt it around earnings season. It works best on **mid and small-cap stocks** where fewer eyes are on the name, and the drift isn't arbitrated away immediately. Large-cap stocks also exhibit PEAD but to a lesser extent or over a shorter horizon (sometimes much of the drift happens within days). A practical approach for an individual: - Use a stock screener each earnings season to find companies with earnings surprise > +X% or SUE in top decile. - Buy a diversified basket of, say, 20 of these stocks the day after earnings, hold for 60–90 days. - Similarly, avoid or short the worst performers. (*Shorting can be risky; one could alternatively buy put options or just focus on long side which has positive expectancy.*)

There isn't a dedicated PEAD ETF, but one could mimic it with sector ETFs if broad sectors have surprise trends (less common). Generally, this is a **stock-picking strategy** around a recurring event. It requires timely data on earnings and expectations (many use the I/B/E/S or Bloomberg estimates). One must be cautious of transaction costs and bid-ask spreads, especially in smaller stocks – these can eat into the drift profits if not managed.

**Bayesian Update (GC + PEAD):** Let  $E$  be the event of a **positive earnings surprise** signal. By itself, a top-quartile earnings surprise stock has a high chance of rising in the next months – say  $P(\text{Success}|E) \approx 0.70$  (as noted, a substantial majority often drift up). Now consider if that stock also triggers a Golden Cross (perhaps the positive earnings caused the price to spike above moving averages, creating a GC shortly after). This dual signal would be very powerful. We might assume that if a stock ultimately performs well (Success), it's quite likely it had good earnings news (let's say  $P(E|\text{Success}, GC) = 0.7$ ), whereas if it failed despite a GC, it probably did not have a positive earnings catalyst ( $P(E|\text{Failure}, GC) = 0.1$ ). Plugging in:

$$P(\text{Success} \mid GC \wedge E) = \frac{0.70 \times 0.79}{0.70 \times 0.79 + 0.10 \times 0.21} \approx 0.97.$$

This yields an **extremely high ~97%** success probability – essentially near-certain. Indeed, a Golden Cross occurring in concert with a strong earnings beat is like a one-two punch of bullishness: the **fundamentals justify** the price increase, and the technical trend confirms **broad momentum**. Historically, one would rarely see a stock fail soon after such a combination (barring external shocks). Thus, using Bayes' rule we update our confidence almost to certainty. Practically, this means if you restrict your buys to instances where a stock has both a bullish technical breakout *and* a stellar earnings report, your batting average (success rate) will be extraordinarily high. (The trade-off is you'll get fewer signals, but those you do get are gold-plated.)

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**Sources:** The strategies above are grounded in extensive research and backtesting. We have cited peer-reviewed studies and reputable strategy tests to support each strategy's performance statistics. For instance, the Golden Cross backtest on S&P 500, momentum strategy performance on Nasdaq and S&P <sup>3</sup>, academic factor data on value and quality, low-volatility index results, and earnings drift studies <sup>14</sup> all reinforce the credibility of these strategies. Each strategy increases the probability of success by exploiting a different market inefficiency or tendency – and when combined with one another (or with a baseline trend

signal), they produce even higher likelihoods of positive outcomes, as demonstrated by the Bayesian updates. Investors should note that **no strategy is foolproof** and all can have periods of underperformance; prudent risk management and diversification remain important. Nonetheless, these five strategies have shown **statistically significant positive results** over long time horizons, giving traders a probabilistic edge in the markets.

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